

Meeting Summaries of October 23rd 2025

Date: October 23rd, 2025

Time: 8:00–8:45 PM(PDT)

Attendees: Dr. Jan, Dr.Jabed, Dr.Yue and Feng

Topic: Feng's report about the work done last week and plan for next week

Summary of Discussion

- Feng gave a brief summary of the work done last week, including:
 1. Completed and submitted proposal.
 2. The solutions to project management, the logic and benefits of using version control system(Git and GitHub) to control the code and all files.
 3. A quick theretical support for making weighted PCA by scaling(weighting) the data matrix before doing PCA computation.
- Dr. Jabed advised Feng to come up with a metric and it should act like a loss function to evaluate the performance of "selecting good reference sites" under different settings of weights.

My thinkings about it: It remains unclear about how to define "good reference sites" are selected in a specific weight setting. Here the attention should be paid on:

- Are we focusing only on the grouping/partitioning of the sites from the weighted PCA results?
- If so, how the individual site contributes to the grouping results? (If the reference group stays stable with same sites, but the relative positions of sites within group change, does it matter or not?)
- Am I sure that the algorithm is working as I expected? What I expected here is that:

In computation stage of this method:

If I set a higher weight on raw stressor "Al", then through the process of: PCA computation → PCs selection → score calculation(based on selected PCs), reference sites and degraded sites are selected.

To the computed results, I expect:

Across these groups, there should be lower mean of Al in the reference group and higher mean of Al in the degraded group.

It should be the most fundamental requirement for this method to work properly, and to achieve this, it is not only dependent on the weight setting, but also how the PCs are selected and how the scores are calculated.

- After the last point, the algorithm behaves always as I expected (higher mean differences of weighted stressors between reference and degraded groups), then I can step out of the computation scope and think about the ecological meanings and its connection to next tasks(clustering, discriminant analysis, etc). On top of this view, I can think about "what is a good metric to evaluate my pollution assessment results(not only the PCA or grouping results)".

- Dr. Jan suggested Feng to consider the toxicity of these stressors and the relationships among them. The concerns of stressor toxicity and their relationships can guide and should guide the weight settings and the PCs selection in the above process. Additionally, selecting the PCs by loadings is not only for pollution assessment, but also for understanding the relationships among stressors and inversely guide the weight settings.
- Dr. Yue mentioned the "functional equivalence (or functional redundancy)", which might be useful and worth exploring in the work.

Decisions Made

- The theoretical proof is not bad, but it needs more focus on the practical implementation ecological meanings, and the statistical evaluation metrics support.

And this report should not be too detailed or technically focused, clear information convey is preferred (I personally think).

Action Items

- Feng will use Git and GitHub to manage the project clearly and stable, the completed work should always be reproduced easily and the codebase should be organized well.
- Feng will finish the code implementation of the talked method, and on top of that, the ecological application meanings and the comprehensive evaluation metric can be explored in a more overall view.

Follow-Up

Temporary meeting time: October 30th, 2025, 20:00–20:45 PM(PDT)

Supportive materials

- A meeting recording with subtitles has been restored locally.